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System and method for verifying user interactions in an extended reality environment

Abstract

A method includes rendering, on displays of an extended reality (XR) device, an XR environment, detecting a sequence of user interactions with an XR application, and identifying, based on the sequence of user interactions with the XR application and the user identification, a historical user interaction data associated with a user. The method further includes generating a first trust token based on the identified historical user interaction data, and receiving an interaction to finalize execution of the sequence of user interactions. In response, the method includes accessing a second trust token associated with the XR application, comparing the first trust token with the second trust token, validating the second trust token based on the comparison, validating each user interaction of the sequence of user interactions based on the identified historical user interaction data, and finalizing execution of the sequence of user interactions in response to the validations.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
8527640	12/2012	Reisman	N/A	N/A
10164985	12/2017	Turgeman	N/A	N/A
10395018	12/2018	Turgeman	N/A	N/A
10586036	12/2019	Turgeman	N/A	N/A
10904246	12/2020	Chari	N/A	N/A
11049082	12/2020	Rice	N/A	N/A
11151234	12/2020	Kontsevich et al.	N/A	N/A
11330012	12/2021	Turgeman	N/A	N/A
11470161	12/2021	Senftleber et al.	N/A	N/A
11546331	12/2022	Senftleber et al.	N/A	N/A
11601398	12/2022	Senftleber et al.	N/A	N/A
11657815	12/2022	Tran et al.	N/A	N/A
2017/0346851	12/2016	Drake	N/A	N/A
2018/0115551	12/2017	Cole	N/A	N/A
2018/0129276	12/2017	Nguyen et al.	N/A	N/A
2021/0226952	12/2020	Senftleber et al.	N/A	N/A
2021/0279695	12/2020	Rice	N/A	N/A
2022/0050983	12/2021	Spivack	N/A	N/A
2022/0247678	12/2021	Atwal et al.	N/A	N/A
2023/0360007	12/2022	Krishnaswamy	N/A	G06Q 20/389
2023/0403152	12/2022	Feijoo	N/A	H04L 9/0894
2024/0143721	12/2023	Naik	N/A	G06T 19/006
2024/0211910	12/2023	Lal	N/A	G06Q 40/04

OTHER PUBLICATIONS

Deb Radcliff; “The metaverse brings new breed of threats to challenge privacy and security gatekeepers;” CSO; <https://www.csoononline.com/article/3686052/the-metaverse-brings-a-new-breed-of-threats-to-challenge-privacy-and-security-gatekeepers.html>; Jan. 23, 2023. cited by applicant

Security; “9 security threats in the metaverse;” <https://www.securitymagazine.com/articles/98142-9-security-threats-in-the-metaverse>; Aug. 10, 2022. cited by applicant

Ashwin Krishnan; “Top metaverse cybersecurity challenges: How to address them;” Tech Target; <https://www.techtarget.com/searchsecurity/tip/Top-metaverse-cybersecurity-challenges-to-consider>; Nov. 18, 2022. cited by applicant

Ramprakash Ramamoorthy, Zoho Corporation; “Why the metaverse is filled with security, privacy

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Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates generally to extended reality environments, and, more specifically, to a system and method for verifying user interactions in an extended reality environment.

BACKGROUND

(2) Some institutions may provide applications suitable for extended reality (XR) environments (e.g., virtual-reality (VR) environments, augmented-reality (AR) environments, mixed-reality (MR) environments, and so forth), which may allow users to perform interactions in XR. As the number of users and associated user avatars interacting with such applications increases, users may perform sensitive user interactions and exchange sensitive data within XR environments. However, many existing XR security processes rely primarily on the service executing and hosting an XR environment, and thus securing particular applications executing within the XR environment and the user interactions therewith may be impeded.

SUMMARY

(3) The system and methods implemented by the system as disclosed in the present disclosure provide technical solutions to the technical problems discussed above by providing systems and methods for verifying and authenticating user interactions within extended reality (XR) environments. The disclosed system and methods provide several practical applications and technical advantages.

(4) The present embodiments are directed to systems and methods for verifying and authenticating user interactions within XR environments. In particular embodiments, one or more processors of a computing system may render, on one or more displays of an XR device, an XR environment. For example, in one embodiment, the XR environment may be configured to facilitate user interactions with the plurality of XR applications while executing within the XR environment. In particular embodiments, the one or more processors may then detect, based on sensor data obtained from one or more sensors of the XR device, a sequence of user interactions with at least one XR application of the plurality of XR applications.

(5) In particular embodiments, the one or more processors may then identify, based at least in part upon the sequence of user interactions with the at least one XR application and a user identification, a historical user interaction data associated with the user. For example, in particular embodiments, the one or more processors may be configured to identify the historical user interaction data by deriving a relational graph based on the sequence of user interactions with the at least one XR application and the user identification. In particular embodiments, the derived relational graph may include a plurality of nodes each corresponding to a respective one of the sequence of user interactions with the at least one XR application, in which a final node of the derived graph relation corresponds to the interaction to finalize execution of the sequence of user interactions. In particular embodiments, the identified historical user interaction data may include one or more of a head pose pattern of the user, an eye gaze pattern of the user, a hand gesture pattern of the user, a body movement pattern of the user, or a clickstream of the user.

(6) In particular embodiments, the one or more processors may then generate a first trust token

based on the identified historical user interaction data. For example, in particular embodiments, the one or more processors may determine, based on the sensor data, one or more device-specific patterns associated with the XR device, and generate the first trust token based on the identified historical user interaction data and the one or more device-specific patterns. In particular embodiments, the one or more processors may then receive an interaction to finalize execution of the sequence of user interactions. In particular embodiments, in response to receiving the interaction to finalize execution of the sequence of user interactions, the one or more processors may then access a second trust token associated with the at least one XR application and compare the first trust token with the second trust token.

(7) For example, in particular embodiments, the one or more processors may receive, from the at least one XR application, the second trust token, which may include an access token configured to permit access to an application programming interface (API) associated with the at least one XR application. The one or more processors may then validate the second trust token based at least in part upon the comparison of the first trust token with the second trust token. For example, in one embodiment, the one or more processors may validate the second trust token based at least in part upon whether the first trust token matches to the second trust token. In particular embodiments, the one or more processors may then validate each user interaction of the sequence of user interactions based at least in part upon the identified historical user interaction data.

(8) In particular embodiments, in response to validating the second trust token and the user interactions of the sequence of user interactions, the one or more processors may then finalize execution of the sequence of user interactions. For example, in particular embodiments, the one or more processors may be configured to finalize execution of the sequence of user interactions by causing the at least one XR application to execute a predetermined action.

(9) The disclosed system and method provide the additional practical application of improving interoperability and security of extended reality (XR) systems (e.g., virtual reality (VR) systems, augmented reality (AR) systems, mixed (MR) systems, and so forth) by way of providing a multi-factor authentication (MFA) process suitable specifically for XR environments. In this way, sensitive information may be seamlessly and securely shared between XR environments and XR applications executing within the XR environments to implement data security, verification and authentication of user interactions, access to an extended reality environment (e.g., metaverse environment, VR environment, AR environment, MR environment, or some combination thereof) and/or XR applications, access to entities within the XR environment, and/or other data interactions performed in both real-world and XR environments.

(10) For example, in accordance with the presently disclosed embodiments, an XR service layer may allow a user wearing an XR device to perform an initial XR service authentication upon launch of an XR environment on the XR device. The XR service layer may then track (e.g., in the background) sequences of user interactions as the user interacts with various XR applications within the XR environment and identify a historical user interaction pattern and a service trust token based thereon. In particular embodiments, upon determining that the user intends to finalize an execution of a particular sequence of user interactions with a particular XR application, the XR service layer may request an access trust token from the particular XR application for comparison with the service trust token in order to verify and authenticate the user with respect to the particular XR application.

(11) Upon verifying and authenticating the user with respect to the particular XR application, the XR service layer may then generate and verify a navigation trust score for the user by comparing each user interaction of the particular sequence of user interactions with the particular XR application with the identified historical user interaction pattern to verify and authenticate that this particular user is in fact the user that performed the particular sequence of user interactions. The XR service layer may then allow finalization of the execution of the particular sequence of user interactions with the particular XR application only when the XR service layer has determined that

each user interaction of the particular sequence of user interactions with the particular XR application matches to the identified historical user interaction pattern. Thus, the disclosed system and methods improve XR systems by providing a verification and authentication process specifically suited for XR environments to secure sensitive data and user interactions exchanged and/or executed within XR environments.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) For a more complete understanding of this disclosure, reference is now made to the following brief description, taken in connection with the accompanying drawings and detailed description, wherein like reference numerals represent like parts.
- (2) FIG. 1 is a block diagram of a system of an extended reality (XR) system and network, in accordance with certain aspects of the present disclosure;
- (3) FIG. 2 is an embodiment of a workflow diagram of an extended reality (XR) system for verifying and authenticating user interactions within XR environments, in accordance with certain aspects of the present disclosure;
- (4) FIGS. 3A and 3B illustrate a diagram of user interaction points and a diagram of relational graphs of user interactions, respectively, in accordance with certain aspects of the present disclosure; and
- (5) FIG. 4 illustrates a flowchart of an example method for verifying and authenticating user interactions within extended reality (XR) environments, in accordance with one or more embodiments of the present disclosure.

DETAILED DESCRIPTION

(6) Example System

(7) FIG. 1 is a block diagram of an extended reality (XR) system and network **100**, in accordance with certain aspects of the present disclosure. In particular embodiments, the system and network **100** may include a first XR device **104**, a second XR device **106**, real-world server **130**, and a XR system **140**, each connected to a network **195**. A first user **110** is associated with the first XR device **104** and a second user **112** is associated with the second XR device **106**. The system and network **100** may be communicatively coupled to the network **195** and may be operable to transmit data between each one of the first XR device **104**, the second XR device **106**, and the XR system **140** through the network **195**.

(8) In particular embodiments, the system and network **100** may improve interoperability and security of extended reality (XR) systems (e.g., virtual reality (VR) systems, augmented reality (AR) systems, mixed (MR) systems, and so forth) so that information may be seamlessly and securely shared between these systems to implement data security, authorization and authentication of data interactions, access to an extended reality environment **102** (e.g., metaverse environment, VR environment, AR environment, MR environment, or some combination thereof), access to entities within the extended reality environment **102** and other data interactions performed in real-world and extended reality environments. For example, user information or sensor data retrieved from a user and/or a user's XR device in a real-world environment may be used in the extended reality environment **102** to determine whether to restrict or allow access to a particular XR application **103** or one or more particular rendered objects associated with the particular XR application **103** within the extended reality environment **102** and/or to perform any kind of action or interaction with the particular XR application **103** or the one or more particular rendered objects associated with the particular XR application **103**.

(9) Additionally, or alternatively, user information collected from the first user **110** and/or assigned to the first user **110** in the real-world environment or extended reality environment **102** may be

used in the extended reality environment **102** to provide the first user **110** access to products, services and/or experiences within the extended reality environment **102**. This process provides improved information security because it authenticates that a first avatar **114** is associated with the first user **110**, not an unauthorized party, and that the first user **110** is authorized to access the extended reality environment **102**, obtain products, services and experiences within the extended reality environment **102** and perform data interactions. Further, a user may obtain a product, service or experience by transferring real-world data objects between real-world entities based on data interactions performed in the extended reality environment **102**.

(10) In particular embodiments, the first user **110** may access the extended reality environment **102** through the first XR device **104**. The first XR device **104** is configured to display a two-dimensional (2D) or three-dimensional (3D) representation of the extended reality environment **102** to the first user **110**. Examples of an extended reality environment **102** may include, but are not limited to, a graphical or virtual representation of a metaverse, a map, a building interior, a landscape, a fictional location, an alternate reality, or any other suitable type of location or environment. The extended reality environment **102** may be configured to use realistic or non-realistic physics for the motion of objects and allow the avatars **114**, **116** to interact with one or more XR applications **103** within the extended reality environment **102**. For example, some extended reality environments **102** may be configured to use gravity whereas other extended reality environments **102** may not be configured to use gravity. Within the extended reality environment **102**, each user may be associated with an avatar (such as the first avatar **114** for the first user **110**). An avatar is a graphical representation of a user at a virtual location within the extended reality environment **102**.

(11) In particular embodiments, the virtual location of each avatar **114**, **116** may be correlated to the physical location of each respective user **110**, **112** in the real-world environment. Examples of avatars **114**, **116** may include, but are not limited to, a person, an animal, or an object. In some embodiments, the features and characteristics of the avatars **114**, **116** may be customizable, and user defined. For example, the size, shape, color, attire, accessories, or any other suitable type of appearance features may be specified by a user. By using the avatars **114**, **116**, the respective users **110**, **112** may be able to move within the extended reality environment **102** to interact with one or more avatars and objects within the extended reality environment **102** while independently remaining at a physical location in the real-world environment or being in transit in the real-world environment.

(12) While engaging in the extended reality environment **102** via the first avatar **114**, the first user **110** may interact with a number of other users, objects and/or entities through a respective avatar. For example, the second user **112** may attempt to engage in an interaction session with the first avatar **114** through a second avatar **116** associated with the second user **112**. In another example, the first avatar **114** of the first user **110** may access an extended reality sub-environment (not shown) within the extended reality environment **102** and perform virtual data interactions within the virtual sub-environment. In the real-world environment, the second user **112** may be physically located at a distance away from the first user **110**. The second user **112** may access the extended reality environment **102** through the second XR device **106** to control the second avatar **116** and attempt to engage in an interaction session with the first user **110** through the first avatar **114**.

(13) Before the interaction between the first avatar **114** and the second avatar **116** occurs, the XR system **140** may authenticate that the first avatar **114** is associated with the first user **110** and not an unauthorized third-party. For example, the first user **110** may be required to sign into a secure portal that provides access to a data file associated with the first user **110**. In some examples, a real-world data file of the first user **110** and a first virtual data file of the first user **110** may be stored and managed by the XR system **140**. Similarly, a second virtual data file associated with the second user **112** is managed by the XR system **140**.

(14) In particular embodiments, the XR system **140** may store other information related to the first

user **110** including, but not limited to, users' profile **162**, account information (e.g., including identity and other details relating to users **110**, **112**), avatar information, digital assets information, or any other suitable type of information that is associated with a user within the extended reality environment **102** and/or the real-world environment. As depicted in FIG. **1**, the XR system **140** may include a processor **150** and a memory **160**. The processor **150** may include one or more processors operably coupled to the memory **160**. In some embodiments, the processor **150** may be any electronic circuitry including, but not limited to, state machines, one or more central processing unit (CPU) chips, logic units, cores (e.g., a multi-core processor), field-programmable gate array (FPGAs), application specific integrated circuits (ASICs), or digital signal processors (DSPs). In other embodiments, the processor **150** may be a programmable logic device, a microcontroller, a microprocessor, or any suitable combination of the preceding.

(15) In particular embodiments, the processor **150** is communicatively coupled to and in signal communication with the memory **160**. The processor **150** may be configured to process data and may be implemented in hardware or software. For example, the processor **150** may be 8-bit, 16-bit, 32-bit, 64-bit or of any other suitable architecture. The processor **150** may include an arithmetic logic unit (ALU) for performing arithmetic and logic operations, processor registers that supply operands to the ALU and store the results of ALU operations, and a control unit that fetches instructions **167** from memory **160** and executes them by directing the coordinated operations of the ALU, registers and other components.

(16) In particular embodiments, the processor **150** may be configured to implement various instructions **167**. For example, the processor **150** may be configured to execute the instructions **167** to implement the XR system **140**. In this way, processor **150** may be a special-purpose computer designed to implement the functions disclosed herein. In particular embodiments, the XR system **140** is implemented using logic units, FPGAs, ASICs, DSPs, or any other suitable hardware. The XR system **140** is configured to operate as described below with reference to FIGS. **2** and **4**, for example. For example, the processor **150** may be configured to perform at least a portion of the method **400** as described in FIG. **4**. In particular embodiments, as will be discussed in greater detail below, the processor **150** may include a user layer **152**, an application layer **154**, and an XR service layer **156**.

(17) In particular embodiments, the memory **160** may include one or more disks, tape drives, or solid-state drives, and may be used as an over-flow data storage device, to store programs when such programs are selected for execution, and to store instructions **167** and data that are read during program execution. The memory **160** may be volatile or non-volatile and may comprise a read-only memory (ROM), random-access memory (RAM), ternary content-addressable memory (TCAM), dynamic random-access memory (DRAM), and static random-access memory (SRAM). In particular embodiments, the memory **160** is operable to store users' profiles **162**, a first user's profile **164**, an authority level **166**, image data **172**, content **174**, a sensitivity level **176**, and a proximity threshold dataset **178**. In particular embodiments, the image data **172** may include any pixel data or voxel data that may be utilized to render and display the extended reality environment **102** (including XR application **103** and avatars **114**, **116**) onto respective displays of the XR devices **104** and **106** of the first user **110** and the second user **112**, respectively.

(18) In particular embodiments, network **195** may include all or a portion of a local area network (LAN), a wide area network (WAN), an overlay network, a software-defined network (SDN), a virtual private network (VPN), a packet data network (e.g., the Internet), a mobile telephone network (e.g., cellular networks, such as 4G or 5G), a Plain Old Telephone (POT) network, a wireless data network (e.g., WiFi, WI Gig, WiMax, etc.), a Long Term Evolution (LTE) network, a Universal Mobile Telecommunications System (UMTS) network, a peer-to-peer (P2P) network, a Bluetooth network, a Near Field Communication network, a Zigbee network, and/or any other suitable network, operable to facilitate communication between the components of system and network **100**. In other embodiments, system and network **100** may not have all of these components

and/or may have other elements instead of, or in addition to, those above.

(19) While the present embodiments may be discussed herein primarily with respect to XR devices **104, 106** being suitable for rendering and displaying the extended reality environment **102** (including XR application **103** and avatars **114, 116**), it should be appreciated that the XR devices **104, 106** may be any user computing devices configured to communicate with other devices, such as a server (e.g., XR system **140**), databases, etc. through the network **195**. Each of the user devices may be configured to perform specific functions described herein and interact with the XR system **140**, e.g., via respective user interfaces. Each of the XR devices **104, 106** is a hardware device that is generally configured to provide hardware and software resources to the first user **110** and the second user **112**, respectively.

(20) Examples of the XR devices **104, 106** include, but are not limited to, a VR device, an AR device, an MR device, a laptop, a computer, a smartphone, a tablet, a smart device, an Internet-of-Things (IoT) device, or some combination thereof. In particular embodiments, the XR devices **104, 106** may each include one or more displays, a touchscreen, a touchpad, keys, buttons, a mouse, or any other suitable type of hardware that allows the respective users **110, 112** to view data and/or to provide inputs into the XR devices **104, 106**. In particular embodiments, the XR devices **104, 106** may also each include any number of sensors suitable for detecting and tracking sensor data (e.g., telemetry data) associated with one or more of the XR devices **104, 106**, the users **110, 112**, the avatars **114, 116**, and/or the one or more XR applications **103**.

(21) For example, in particular embodiments, the number of sensors may include one or more of inertial measurement units (IMUs), one or more monochromatic cameras, one or more visible-light cameras (VLCs), one or more infrared (IR) cameras, one or more depth cameras, one or more accelerometers, one or more magnetometers, one or more gyroscopes, or other sensors that may be suitable for detecting and tracking a head pose of the respective users **110, 112**, an eye gaze of the respective users **110, 112**, a hand gesture of the respective users **110, 112**, a face of the respective users **110, 112**, a body movement of the respective users **110, 112**, a haptic control of the respective users **110, 112**, a spatial proximity of the avatars **114, 116** with respect to one or more rendered objects associated with the XR application **103**, an object pose of one or more rendered objects associated with the XR application **103**.

(22) In particular embodiments, as previously noted, the processor **150** may include the user layer **152**. The user layer **152** may include any software layer or software system (e.g., user interfaces (UIs), graphical user interfaces (GUIs), buttons, menus, text boxes, widgets, objects, event handlers, macros, scripts, and so forth) suitable for receiving and processing interactions and/or inputs of the user **110**. In particular embodiments, the application layer **154** may include any software layer, middleware layer, or software or middleware system suitable for servicing and processing communications, data transfers, network protocols, and authentication and authorization data between any number of XR applications **103** executing within the XR environment **102** as executed by the XR device **104**.

(23) In particular embodiments, the XR service layer **156** may include any software layer, middleware layer, or software or middleware system (e.g., operating system, edge operating system, and so forth) suitable for hosting and servicing the XR environment **102** as executed by the XR devices **104, 106**. For example, in one embodiment, the XR service layer **156** may include one or more of a Platform as a Service (PaaS) layer, a Software as a Service (SaaS) layer, an Infrastructure as a Service (IaaS) layer, a Compute as a Service (CaaS) layer, a Data as a Service (DaaS) layer, a Database as a Service (DBaaS) layer, or other similar cloud-based computing architecture (e.g., “X” as a Service (XaaS)) suitable for hosting and servicing the XR environment **102**.

(24) In particular embodiments, the XR service layer **156** may also be suitable for performing one or more machine-learning based algorithms. For example, the XR service layer **156** may be configured to identify the users **110, 112** using any suitable face-tracking algorithm, eye-tracking

algorithm, gesture recognition algorithm, and so forth. In one embodiment, the XR service layer **156** identifies the users **110, 112** by searching for each users' **110** image in the user profile **162** in memory **160**, and for each of the users **110, 112**, matching users' **110, 112** image to one of the stored user's profiles **164**. In particular embodiments, the users' profile **162** in memory **160** includes users' authority level **166** that indicates which content **174** containing sensitive information, the user **110, 112** are authorized to view.

(25) For example, a first viewer among users **110, 112** with a low authority level **166** is authorized to view content **174** with a low sensitivity level **176**. A second viewer among users **110, 112** with a high authority level **166** is authorized to view content **174** with low to high sensitivity levels **176**. The authority level **166** of each of the users **110, 112** may be stored in their respective user's profile **164**. Once the XR service layer **156** determines the identity of users **110, 112**, the XR service layer **156** searches the users' profile **162** and matches the users **110, 112** image to one of the user's profiles **164**.

(26) In particular embodiments, the proximity threshold dataset **178** may also include tables for different content **174**, different item sensitivity levels **176**, different XR device **106** and/or second user avatar **116** virtual locations, and/or the like. For example, a first table may include distances from XR device **106** and/or second user avatar **116**, from which the same image with different virtual locations is identifiable. A second table may include distances from XR device **106** and/or second user avatar **116**, from which the same text with different font virtual locations is identifiable. A third table may include distances from XR device **106** and/or second user avatar **116**, from which the same image with the same virtual location but different resolutions is identifiable. A fourth table may include distances from the XR device **106** and/or second user avatar **116** with different virtual locations, from which the same image is identifiable. Then, based on these data points, a proximity threshold **117** for content **174** that has the same or closest data point to a particular datapoint may be determined.

(27) Verifying and Authenticating User Interactions in Extended Reality Environments

(28) Embodiments of the present disclosure discuss techniques system for verifying and authenticating user interactions within extended reality (XR) environments.

(29) FIG. 2 is an embodiment of a workflow diagram **200** of an extended reality (XR) system for verifying and authenticating user interactions within XR environments, in accordance with certain aspects of the present disclosure. In particular embodiments, the workflow diagram **200** may be performed by the XR service layer **156** and/or a combination of the XR service layer **156** and the application layer **154** as executed by the processor **150** discussed above with respect to FIG. 1. In particular embodiments, the workflow diagram **200** may begin with the XR service layer **156** allowing the user **110** wearing the XR device **104** to perform an initial XR service authentication **202** upon launch of the XR environment **102** on the XR device **104**. For example, in one embodiment, the initial XR service authentication **202** may include a multifactor authentication (MFA) in which the user **110** may input one or more identification data (e.g., user identification data, personal identification data, user avatar identification data) using the XR device **104** and/or mobile device **104**.

(30) In particular embodiments, the workflow diagram **200** may further include the XR service layer **156** tracking and analyzing (e.g., in the background) user interactions **206** as the user **110** interacts with the XR application **103** executing within the XR environment **103**. In particular embodiments, the user interactions **206** may include, for example, one or more user inputs (e.g., eye-gaze input, face expression input, tactile input, a head pose input, a hand gesture, a full-body gesture, a voice input, biometric input, and so forth) suitable for conveying a user intent with respect to the XR application **103**. In particular embodiments, the workflow diagram **200** may further include the XR service layer **156** extracting device data **208** based on sensor data obtained by one or more sensors of the XR device **104** as the user **110** interacts with the XR application **103**.

(31) For example, in particular embodiments, the extracted device data **208** may include an

indication of one or more device capabilities of the XR device **104**, such as an eye-tracking capability, a face-tracking capability, a haptic feedback capability, a head pose tracking capability, a hand tracking capability, a full-body tracking capability, a voice control capability, a biometric input capability, or other device data that may be associated with the particular XR device **104**. In particular embodiments, the workflow diagram **200** may then continue with the XR service layer **156** utilizing a data analyzer **210** to identify a device-specific pattern **212** based on the extracted device data **208** and a historical user interaction navigation pattern **218** based on the extracted device data **208** and raw user interactions **220** (e.g., visual data representative of individual user selections, user clicks, user points, user gestures, and the respective locations thereof).

(32) In particular embodiments, the device-specific pattern **212** may include, for example, one or more patterns associated with the user input modality (e.g., eye-gaze input modality, face expression input modality, tactile input modality, a head pose input modality, a hand gesture input modality, a full-body gesture input modality, voice input modality, biometric input modality, and so forth) utilized by the user **110** to perform user interactions with respect to the identified device capabilities of the XR device **104**.

(33) Similarly, in particular embodiments, the historical user interaction navigation pattern **218** may include, for example, one or more patterns associated with the manner in which the user **110** performs a sequence of user interactions **206** (e.g., a head pose pattern of the user **110** while performing a sequence of user interactions **206**, an eye gaze pattern of the user **110** while performing a sequence of user interactions **206**, a hand gesture pattern of the user **110** while performing a sequence of user interactions **206**, a full-body movement pattern of the user **110** while performing a sequence of user interactions **206**, a clickstream of the user **110** while performing a sequence of user interactions **206**, and so forth) with respect to the XR application **103**.

(34) For example, in particular embodiments, the historical user interaction navigation pattern **218** may be identified by deriving a relational graph **222** representative of a pattern structure **224** of the sequence of user interactions **206** with the XR application **103**. In particular embodiments, as will be further described below by FIGS. 3A and 3B, the derived relational graph **222** may include any graph structure, which includes a number of individual nodes each corresponding to a respective one of the sequence of user interactions **206** with the XR application **103**.

(35) In particular embodiments, the workflow diagram **200** may then include the XR service layer **156** generating a service trust token **214** based on the device-specific pattern **212** and the historical user interaction navigation pattern **218**. In particular embodiments, the workflow diagram **200** may then include the XR service layer **156** storing the service trust token **214** and the derived relational graph **222** to a service database **216**. In particular embodiments, upon determining that the user **110** intends to finalize an execution of a particular sequence of user interactions **206** with the XR application **103** based on the user's performance of a final interaction **242**, the workflow diagram **200** may include the XR service layer **156** requesting an access trust token **231** associated with the XR application **103** from an application server **230** and application database **232**. For example, in particular embodiments, the access trust token **231** may include any access token (e.g., user authentication credential) suitable for permitting user **110** access to an application programming interface (API) associated with the XR application **103** to perform one or more specific actions.

(36) In particular embodiments, the workflow diagram **200** may include the XR service layer **156** comparing the service trust token **214** and the access trust token **231** in order to verify and authenticate the user **110** with respect to the XR application **103** via an authentication server **226** running on the XR service layer **156** and/or an authentication client **228** running on the application layer **154**. In particular embodiments, the workflow diagram **200** may then include the XR service layer **156** verifying and authenticating the user **110** with respect to the XR application **103** by determining (at validate trust token functional block **236**) whether the service trust token **214** matches to the access trust token **231** by way of an authenticator **234**.

(37) In particular embodiments, in response to determining (at validate trust token functional block

236) that the service trust token **214** does not match to the access trust token **231**, the workflow diagram **200** may include the XR service layer **156** rejecting (at rejection functional block **238**) the finalization of the execution of the sequence of user interactions **206** with the XR application **103**. On the other hand, in response to determining (at validate trust token functional block **236**) that the service trust token **214** matches to the access trust token **231**, the workflow diagram **200** may then include the XR service layer **156** generating and verifying a navigation trust score (at validate trust token functional block **240**) for the user **110**.

(38) For example, in particular embodiments, the XR service layer **156** may generate and verify the navigation trust score (at validate trust token functional block **240**) for the user **110** by comparing each user interaction of the sequence of user interactions **206** with the XR application **103** with the identified the historical user interaction navigation pattern **218** to verify and authenticate, for example, that this particular user **110** is in fact the user **110** that performed the particular sequence of user interactions **206**. In particular embodiments, in response to determining that each user interaction of the sequence of user interactions **206** with the XR application **103** does not match to, or accord with, the identified the historical user interaction navigation pattern **218** based on the navigation trust score (at validate trust token functional block **240**), the workflow diagram **200** may include the XR service layer **156** rejecting (at rejection functional block **238**) the finalization of the execution of the sequence of user interactions **206** with the XR application **103**.

(39) On the hand, in response to determining that each user interaction of the sequence of user interactions **206** with the XR application **103** matches to, or accords with, the identified the historical user interaction navigation pattern **218** based on the navigation trust score (at validate trust token functional block **240**), the workflow diagram **200** may include the XR service layer **156** allowing finalization of the execution of the sequence of user interactions **206** with the XR application **103** based on the user's performance of the final interaction **242**. For example, in one embodiment, the finalization of the execution of the sequence of user interactions **206** with the XR application **103** may include, for example, causing the XR application **103** to execute a predetermined action (e.g., launch a web page, compute a result, execute a transmission of data, execute a reception of data, transform an object, alter an object, display a widget, update and redisplay a widget, launch another XR application, change a state, and so forth).

(40) In accordance with the presently disclosed embodiments, by allowing finalization of the execution of the sequence of user interactions **206** with the XR application **103** only when the XR service layer **156** has determined that each user interaction of the sequence of user interactions **206** with the XR application **103** matches to, or accords with, the identified historical user interaction navigation pattern **218**, the present embodiments may improve XR systems by providing a verification and authentication process specifically suited for XR environments to secure sensitive data and user interactions exchanged and/or executed within XR environments.

(41) FIGS. **3A** and **3B** illustrate a diagram of user interaction points **300A** and a diagram of relational graphs of user interactions **300B**, respectively, in accordance with certain aspects of the present disclosure. As depicted by FIG. **3A**, the diagram of user interaction points **300A** may include a first aggregate of interactions **302A** and a second aggregate of interactions **302B**. In particular embodiments, the first aggregate of interactions **302A** and the second aggregate of interactions **302B** may each include, for example, raw visual data representative of individual user selections, user clicks, user points, user gestures, and the respective locations thereof, that may be tracked and recorded as the user **110** performs user interactions with the XR application **103**.

(42) As depicted by FIG. **3B**, the diagram of relational graphs of user interactions **300B** may include a first sequence of user interactions **304A** and a second sequence of user interactions **304B**. For example, in particular embodiments, the first sequence of user interactions **304A** and the second sequence of user interactions **304B** may each include, for example, a derived relational graph representative of a pattern structure of the particular sequences of user interactions with the XR application. For example, as further depicted, the first sequence of user interactions **304A** and

the second sequence of user interactions **304B** may each include a number of individual and sequential nodes (e.g., “1”, “2”, “3”, “4”, and “5”, . . . “N”) each corresponding to a respective user interaction of the sequence of user interactions **304A**, **304B** with the XR application. In accordance with the presently disclosed embodiments, the first sequence of user interactions **304A** and the second sequence of user interactions **304B** may each be tracked until a final node or interaction to finalize execution of the sequence of user interactions **306** is performed by the user **110**. The derived relational graph may then be recorded and stored as being representative of an identified historical user interaction navigation pattern for the user **110** in accordance with the presently disclosed embodiments.

(43) FIG. 4 illustrates a flowchart of an example method **400** for verifying and authenticating user interactions within extended reality (XR) environments, in accordance with one or more embodiments of the present disclosure. The method **400** may be performed by the system and network **100** as described above with respect to FIG. 1. The method **400** may begin at block **402** with the XR system **140** rendering on one or more displays of an XR device **104** an extended reality (XR) environment **102**. For example, in one embodiment, the XR environment **102** may include any XR environment (e.g., VR environment, AR environment, MR environment, and so forth) suitable for facilitating user interactions **206** with the XR application **103** while executing within the XR environment **102**. The method **400** may continue at block **404** with the XR system **140** detecting, based on sensor data obtained from one or more sensors of the XR device **104**, a sequence of user interactions **304A** with the XR application **103**.

(44) In particular embodiments, the method **400** may continue at block **406** with the XR system **140** identifying, based on the sequence of user interactions **304A** with the XR application **103** and a user identification, a historical user interaction data associated with the user **110**. For example, in particular embodiments, the historical user interaction data associated with the user **110** may include the user navigation patterns **218** and the device-specific patterns **212**. In particular embodiments, the method **400** may continue at block **408** with the XR system **140** generating a first trust token **214** based on the identified historical user interaction data associated with the user **110**.

(45) In particular embodiments, the method **400** may continue at decision **410** with the XR system **140** determining whether a final user interaction **306** to finalize the sequence of user interactions **304A** has been received. In particular embodiments, in response to determining that the final user interaction **306** to finalize the sequence of user interactions **304A** has been received, the method **400** may then continue at block **412** with the XR system **140** accessing a second trust token associated with the XR application **103**. For example, in particular embodiments, the XR system **140** may receive, from the XR application **103**, the second trust token **231**, which may include an access token suitable for permitting access to an API associated with the XR application **103**. In response to determining that the final user interaction **306** to finalize the sequence of user interactions **304A** has not been received, the method **400** may return to block **406**.

(46) In particular embodiments, the method **400** may continue at decision **414** with the XR system **140** determining whether the first trust token **214** matches to the second trust token **231**. For example, in particular embodiments, the XR system **140** may validate the second trust token **231** based on a comparison of the first trust token **214** with the second trust token **231** to determine whether the first trust token **214** matches to the second trust token **231**. In particular embodiments, in response to determining that the first trust token **214** does not match to the second trust token **231**, the method **400** may conclude with the XR system **140** rejecting the execution of the sequence of user interactions **304A**. In particular embodiments, in response to determining that the first trust token **214** matches to the second trust token **231**, the method **400** may continue at decision **416** with the XR system **140** determining whether the sequence of user interactions **304A** matches with the identified historical user interaction data associated with the user **110**.

(47) For example, in particular embodiments, the XR system **140** may validate each user interaction of the sequence of user interactions **304A** based on the identified historical user

interaction data associated with the user **110** by comparing each user interaction of the sequence of user interactions **304A** against the identified historical user interaction data associated with the user **110**. In particular embodiments, the method **400** may then conclude at block **418** with the XR system **140** finalizing execution of the sequence of user interactions **304A**. For example, in particular embodiments, the XR system **140** may, in response to the final user interaction **306** to finalize the sequence of user interactions **304A** (e.g., a confirmation interaction), may finalize execution of the sequence of user interactions **304A** by causing the XR application **103** to execute a predetermined action (e.g., launch a web page, compute a result, execute a transmission of data, execute a reception of data, transform an object, alter an object, display a widget, update and redisplay a widget, launch another XR application, change a state, and so forth).

(48) While several embodiments have been provided in the present disclosure, it should be understood that the disclosed systems and methods might be embodied in many other specific forms without departing from the spirit or scope of the present disclosure. The present examples are to be considered as illustrative and not restrictive, and the intention is not to be limited to the details given herein. For example, the various elements or components may be combined or integrated in another system or certain features may be omitted, or not implemented.

(49) In addition, techniques, systems, subsystems, and methods described and illustrated in the various embodiments as discrete or separate may be combined or integrated with other systems, modules, techniques, or methods without departing from the scope of the present disclosure. Other items shown or discussed as coupled or directly coupled or communicating with each other may be indirectly coupled or communicating through some interface, device, or intermediate component whether electrically, mechanically, or otherwise. Other examples of changes, substitutions, and alterations are ascertainable by one skilled in the art and could be made without departing from the spirit and scope disclosed herein.

(50) To aid the Patent Office, and any readers of any patent issued on this application in interpreting the claims appended hereto, applicants note that they do not intend any of the appended claims to invoke 35 U.S.C. § 112 (f) as it exists on the date of filing hereof unless the words “means for” or “step for” are explicitly used in the particular claim.

Claims

1. A system, comprising: a memory configured to store a user identification for a user, image data utilized to render an extended reality (XR) environment on an XR device associated with the user, and a plurality of XR applications executable within the XR environment; and one or more processors operably coupled to the memory and configured to: render, on one or more displays of the XR device, the XR environment, wherein the XR environment is configured to facilitate user interactions with the plurality of XR applications while executing within the XR environment; detect, based on sensor data obtained from one or more sensors of the XR device, a sequence of user interactions with at least one XR application of the plurality of XR applications; identify, based at least in part upon the sequence of user interactions with the at least one XR application and the user identification, a historical user interaction data associated with the user; generate a first trust token based on the identified historical user interaction data; and receive an interaction to finalize execution of the sequence of user interactions, and, in response: access a second trust token associated with the at least one XR application; compare the first trust token with the second trust token; validate the second trust token based at least in part upon the comparison of the first trust token with the second trust token; validate each user interaction of the sequence of user interactions based at least in part upon the identified historical user interaction data; and in response to validating the second trust token and the user interactions of the sequence of user interactions, finalize execution of the sequence of user interactions.

2. The system of claim 1, wherein the one or more processors are configured to identify the

historical user interaction data by deriving a relational graph based on the sequence of user interactions with the at least one XR application and the user identification.

3. The system of claim 2, wherein the derived relational graph comprises a plurality of nodes each corresponding to a respective one of the sequence of user interactions with the at least one XR application, and wherein a final node of the derived relational graph corresponds to the interaction to finalize execution of the sequence of user interactions.

4. The system of claim 1, wherein the one or more processors are further configured to: determine, based on the sensor data, one or more device-specific patterns associated with the XR device; and generate the first trust token based on the identified historical user interaction data and the one or more device-specific patterns.

5. The system of claim 1, wherein the one or more processors are further configured to: receive, from the at least one XR application, the second trust token, wherein the second trust token comprises an access token configured to permit access to an application programming interface (API) associated with the at least one XR application; and validate the second trust token based at least in part upon whether the first trust token matches to the second trust token.

6. The system of claim 1, wherein the one or more processors are configured to finalize execution of the sequence of user interactions by causing the at least one XR application to execute a predetermined action.

7. The system of claim 1, wherein the identified historical user interaction data comprises one or more of a head pose pattern of the user, an eye gaze pattern of the user, a hand gesture pattern of the user, a body movement pattern of the user, or a clickstream of the user.

8. A method, comprising: rendering, on one or more displays of an extended reality (XR) device associated with a user, an XR environment, wherein the XR environment is configured to facilitate user interactions with a plurality of XR applications while executing within the XR environment; detecting, based on sensor data obtained from one or more sensors of the XR device, a sequence of user interactions with at least one XR application of the plurality of XR applications; identifying, based at least in part upon the sequence of user interactions with the at least one XR application and a user identification, a historical user interaction data associated with the user; generating a first trust token based on the identified historical user interaction data; and receiving an interaction to finalize execution of the sequence of user interactions, and, in response: accessing a second trust token associated with the at least one XR application; comparing the first trust token with the second trust token; validating the second trust token based at least in part upon the comparison of the first trust token with the second trust token; validating each user interaction of the sequence of user interactions based at least in part upon the identified historical user interaction data; and in response to validating the second trust token and the user interactions of the sequence of user interactions, finalizing execution of the sequence of user interactions.

9. The method of claim 8, wherein identifying the historical user interaction data comprises deriving a relational graph based on the sequence of user interactions with the at least one XR application and the user identification.

10. The method of claim 9, wherein the derived relational graph comprises a plurality of nodes each corresponding to a respective one of the sequence of user interactions with the at least one XR application, and wherein a final node of the derived relational graph corresponds to the interaction to finalize execution of the sequence of user interactions.

11. The method of claim 8, further comprising: determining, based on the sensor data, one or more device-specific patterns associated with the XR device; and generating the first trust token based on the identified historical user interaction data and the one or more device-specific patterns.

12. The method of claim 8, further comprising: receiving, from the at least one XR application, the second trust token, wherein the second trust token comprises an access token configured to permit access to an application programming interface (API) associated with the at least one XR application; and validating the second trust token based at least in part upon whether the first trust

token matches to the second trust token.

13. The method of claim 8, wherein finalizing execution of the sequence of user interactions comprises causing the at least one XR application to execute a predetermined action.

14. The method of claim 8, wherein the identified historical user interaction data comprises one or more of a head pose pattern of the user, an eye gaze pattern of the user, a hand gesture pattern of the user, a body movement pattern of the user, or a clickstream of the user.

15. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors of a system, cause the one or more processors to: render, on one or more displays of an extended reality (XR) device associated with a user, an XR environment, wherein the XR environment is configured to facilitate user interactions with a plurality of XR applications while executing within the XR environment; detect, based on sensor data obtained from one or more sensors of the XR device, a sequence of user interactions with at least one XR application of the plurality of XR applications; identify, based at least in part upon the sequence of user interactions with the at least one XR application and a user identification, a historical user interaction data associated with the user; generate a first trust token based on the identified historical user interaction data; access a second trust token associating with the at least one XR application based at least in part upon; and receive an interaction to finalize execution of the sequence of user interactions, and, in response: access a second trust token associated with the at least one XR application; compare the first trust token with the second trust token; validate the second trust token based at least in part upon the comparison of the first trust token with the second trust token; validate each user interaction of the sequence of user interactions based at least in part upon the identified historical user interaction data; and in response to validating the second trust token and the user interactions of the sequence of user interactions, finalize execution of the sequence of user interactions.

16. The non-transitory computer-readable medium of claim 15, wherein the instructions to identify the historical user interaction data further cause the one or more processors to derive a relational graph based on the sequence of user interactions with the at least one XR application and the user identification.

17. The non-transitory computer-readable medium of claim 16, wherein the derived relational graph comprises a plurality of nodes each corresponding to a respective one of the sequence of user interactions with the at least one XR application, and wherein a final node of the derived relational graph corresponds to the interaction to finalize execution of the sequence of user interactions.

18. The non-transitory computer-readable medium of claim 15, wherein the instructions further cause the one or more processors to: determine, based on the sensor data, one or more device-specific patterns associated with the XR device; and generate the first trust token based on the identified historical user interaction data and the one or more device-specific patterns.

19. The non-transitory computer-readable medium of claim 15, wherein the instructions further cause the one or more processors to: receive, from the at least one XR application, the second trust token, wherein the second trust token comprises an access token configured to permit access to an application programming interface (API) associated with the at least one XR application; and validate the second trust token based at least in part upon whether the first trust token matches to the second trust token.

20. The non-transitory computer-readable medium of claim 15, wherein the identified historical user interaction data comprises one or more of a head pose pattern of the user, an eye gaze pattern of the user, a hand gesture pattern of the user, a body movement pattern of the user, or a clickstream of the user.
